

Singapore

Competition and Consumer Commission

Analytics Tool for Complaints

The Competition and Consumer Commission of Singapore (“CCCS”) developed an in-house text analytics tool to analyze large volumes of complaints that are received from the public.

The tool assists CCCS in identifying key industries and thematic issues from the complaints. In particular, it enables CCCS to visualize the underlying time-series trends from complaints and to identify and prioritize a list of industries/issues such as beauty services and renovation. From there, CCCS could take further action by conducting market studies to understand industry developments better or follow up with other government agencies and stakeholders to improve the competitive dynamics of the industry. The tool also reduced the duration needed for CCCS to analyze complaints and allowed case teams to complete analyzing a set of the monthly complaints within a single day, as compared to several weeks with manual filtering and analysis. It also improves the objectivity of the results by reducing the dependence on human judgment which could be highly subjective.

The project involved the use of the R programming language, and software R Studio to create the text analytics tool. Data visualization libraries (such as word clouds) were also used to cluster text around key topics.

Document Similarity and Bid-Rigging Detection Tools

As part of the investigation process for potential bid-rigging cases, investigation teams will generally have to go through the tedious process of reviewing numerous proposal documents submitted by companies for past tenders. Such manual processes are time-consuming and are more prone to human error or lapses.

To address these bottlenecks, the Competition and Consumer Commission of Singapore (“CCCS”), in collaboration with the Government Technology Agency, is building a software tool that automates the process of comparing the similarity of documents at the sentence and document level. The tool employs text analytics techniques such as Longest Common Sequence and Bilingual Evaluation Understudy (“BLEU”) to generate similarity scores for sentence and document level comparisons. This allows officers to zoom in on sentences and documents that display high levels of similarity instead of having to comb through large volumes of documents, thereby significantly reducing the time and effort that goes into evidence review, and also minimizing the likelihood of missing out on similar documents through human error.

This document similarity tool can be used in conjunction with CCCS’s Bid Rigging Detection Tool (“BRDT”) to identify bid rigging behavior. The BRDT is a tool developed in-house that analyses bid prices and bid patterns to flag tenders based on a variety of quantitative indicators that signal suspicious bidding behavior. The document similarity tool is then applied to perform a deep dive into the bid documents submitted to the respective tenders.